YARON OGEN, Ph.D.

Remote Sensing Scientist/Consultant · Imaging & Point Spectroscopy · Geospatial Data Modelling

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PROFILE

Remote sensing scientist with a Ph.D. in soil spectroscopy and more than a decade of experience developing quantitative models that link laboratory spectral measurements to airborne and satellite Earth observation data. Author of 15 peer-reviewed publications, predominantly in Q1 journals, spanning soil and mineral spectroscopy, geochemical prediction in mining environments, and environmental applications of Earth observation. Currently leading the scientific and research function of an applied geospatial technology company, translating research-grade methods into operational, Python-based analytical workflows.

PROFESSIONAL EXPERIENCE

Chief Research Officer (co-founder) July 2024 – September 2026

Xsupra GmbH, Halle (Saale), Germany

Responsible for leading the company's scientific and research activities, including the development of data-driven models and methodologies for agricultural applications. Design and implement Python-based algorithms and workflows with a strong focus on satellite image processing, drone imagery, modelling, and geospatial analysis.

Postdoctoral Researcher June 2019 – December 2022

Institute of Geosciences and Geography, Martin Luther University Halle-Wittenberg, Germany

Project: **ADRIANA** (Airborne spectral Detection of Reusable Industry mAterials in tailiNgs fAcilities), funded by the Federal Ministry of Education and Research (BMBF), Germany. Objective: to develop new strategies and techniques to monitor and predict the geochemical composition of tailings in the Cu–Mo mine in Erdenet, Mongolia. Supervisor: Prof. Cornelia Glaesser.

EDUCATION

Ph.D. — Remote Sensing Laboratory November 2018

Department of Geography and the Human Environment & Porter School of Environmental Studies, Tel Aviv University, Israel

Dissertation: *"Soil Spectroscopy in the Visible, Near and Shortwave Infrared (0.4–2.5 µm) for Precise Soil Mapping".* Supervisors: Prof. Eyal Ben-Dor and Dr. Naftaly Goldshleger.

M.A. — Remote Sensing Laboratory July 2012

Geography and the Human Environment, Tel Aviv University, Israel

Thesis: *"Generating DEM from HRS Data to Improve Classical Spectral Mapping".* Supervisor: Prof. Eyal Ben-Dor.

B.A., cum laude July 2007

Geography and the Human Environment, Tel Aviv University, Israel

PEER-REVIEWED PUBLICATIONS

15 peer-reviewed articles. Impact factors and quartiles as reported at time of publication.

Ogen, Y., Denk, M., Glaesser, C., Eichstaedt, H. (2022). A novel method for predicting the geochemical composition of tailings with laboratory field and hyperspectral airborne data using a regression and classification-based approach. *European Journal of Remote Sensing* 55, 453–470.

Ben-Dor, E., Francos, N., **Ogen, Y.**, Banin, A. (2022). Aggregate size distribution of arid and semiarid laboratory soils (<2 mm) as predicted by VIS–NIR–SWIR spectroscopy. *Geoderma* 416, 115819. (IF: 6.114, Q1)

Ogen, Y., Denk, M., Glaesser, C., Eichstaedt, H., Kahnt, R., Loeser, R., Suppes, R., Chimeddorj, M., Tsedenbaljir, T., Alyeksandr, U., et al. (2021). Quantification of the spectral variability of ore-bearing granodiorite under supervised and semisupervised conditions: an upscaling approach. *Journal of Spectroscopy* 2021, e2580827. (IF: 1.914, Q4)

Francos, N., **Ogen, Y.**, Ben-Dor, E. (2021). Spectral assessment of organic matter with different composition using reflectance spectroscopy. *Remote Sensing* 13, 1549. (IF: 4.848, Q1)

Eichstaedt, H., Tsedenbaljir, T., Kahnt, R., Denk, M., **Ogen, Y.**, Glaesser, C., Loeser, R., Suppes, R., Alyeksandr, U., Oyunbuyan, T., et al. (2020). Quantitative estimation of clay minerals in airborne hyperspectral data using a calibration field. *Journal of Applied Remote Sensing* 14, 034524. (IF: 1.530, Q2)

Ogen, Y. (2020). Assessing nitrogen dioxide (NO_2) levels as a contributing factor to coronavirus (COVID-19) fatality. *Science of the Total Environment* 726, 138605. (IF: 7.963, Q1)

Ogen, Y. (2020). Response to the commentary by Alexandra A. Chudnovsky on "Assessing nitrogen dioxide (NO_2) levels as a contributing factor to coronavirus (COVID-19) fatality". *Science of the Total Environment* 139239. (IF: 7.963, Q1)

Ogen, Y. (2020). Response to the commentary by Pisoni, E. and Van Dingenen, R. on "Assessing nitrogen dioxide (NO_2) levels as a contributing factor to coronavirus (COVID-19) fatality". *Science of the Total Environment* 140672. (IF: 7.963, Q1)

Tziolas, N., Tsakiridis, N., **Ogen, Y.**, Kalopesa, E., Ben-Dor, E., Theocharis, J., Zalidis, G. (2020). An integrated methodology using open soil spectral libraries and Earth observation data for soil organic carbon estimations in support of soil-related SDGs. *Remote Sensing of Environment* 111793. (IF: 10.164, Q1)

Ogen, Y., Faigenbaum-Golovin, S., Granot, A., Shkolnisky, Y., Goldshleger, N., Ben-Dor, E. (2019). Removing the moisture effect on soil reflectance properties: a case study of clay-content prediction. *Pedosphere* 29(4), 421–431. (IF: 3.911, Q1)

Ogen, Y., Zaluda, J., Francos, N., Goldshleger, N., Ben-Dor, E. (2019). Cluster-based spectral models for a robust assessment of soil properties. *Geoderma* 340, 175–184. (IF: 6.114, Q1)

Chabrillat, S., Gholizadeh, A., Neumann, C., Berger, D., Milewski, R., **Ogen, Y.**, Ben-Dor, E. (2019). Preparing a soil spectral library using the Internal Soil Standard (ISS) method: influence of extreme different humidity laboratory conditions. *Geoderma* 355, 113855. (IF: 6.114, Q1)

Ogen, Y., Neumann, C., Chabrillat, S., Goldshleger, N., Ben-Dor, E. (2018). Evaluating the detection limit of organic matter using point and imaging spectroscopy. *Geoderma* 321, 100–109. (IF: 6.114, Q1)

Ogen, Y., Goldshleger, N., Ben-Dor, E. (2017). 3D spectral analysis in the VNIR–SWIR spectral region as a tool for soil classification. *Geoderma* 302, 100–110. doi:10.1016/j.geoderma.2017.04.020 (IF: 6.114, Q1)

Notesco, G., **Ogen, Y.**, Ben-Dor, E. (2016). Integration of hyperspectral shortwave and longwave infrared remote-sensing data for mineral mapping of Makhtesh Ramon in Israel. *Remote Sensing* 8, 318. (IF: 4.848, Q1)

MANUSCRIPTS IN PREPARATION

Ogen, Y., Denk, M., Glaesser, C., Eichstaedt, H. Utilizing LiDAR intensity data to improve prediction models of copper and molybdenum generated by HySpex hyperspectral imagery **(in preparation).**

Ogen, Y., Denk, M., Glaesser, C. Determining the spectral detection limit of metals and minerals using the VNIR–SWIR spectral regions under supervised conditions **(in preparation).**

RESEARCH PROJECTS & SCIENTIFIC WORKING GROUPS

- **National Soil Spectral Library (SSL) of Israel** — Project Manager, Remote Sensing Laboratory, Tel Aviv University. Building a national soil spectral library of the soils of Israel, funded by the Ministry of Agriculture, Israel.
- **GEO-CRADLE (EU Horizon 2020)** — Project Manager, Remote Sensing Laboratory, Tel Aviv University. Coordinating and integRating state-of-the-art Earth Observation Activities in the regions of North Africa, Middle East and Balkans, and Developing Links with GEO-related initiatives towards GEOSS. Included full responsibility for organising international workshops.
- **ADRIANA (BMBF, Germany)** — Postdoctoral researcher. Airborne spectral detection of reusable industrial materials in tailings facilities, Erdenet Cu–Mo mine, Mongolia.

TEACHING EXPERIENCE

Lecturer — Department of Geography and the Human Environment, Tel Aviv University. Fully responsible for all aspects of course development, instruction, and assessment.

- Fundamentals of Physics and Chemistry for Environmental Sciences
- Practices and Applications in Image Processing

Teaching Assistant — Department of Geography and the Human Environment, Tel Aviv University.

- Introduction to Climate
- Remote Sensing and Image Processing
- Hyperspectral Remote Sensing
- Soils of Israel

CONFERENCE PRESENTATIONS & WORKSHOPS

- *Identifying copper hot spots in the subsurface of an operational tailings facility using Landsat time-series analysis.* GRSG, December 2022, Enschede, The Netherlands.
- *Utilizing LiDAR intensity data to improve copper and molybdenum prediction models in a high-wetness environment.* EARSeL, June 2022, Potsdam, Germany.
- *A novel method for predicting the geochemical composition of tailings with laboratory field and hyperspectral airborne data using regression- and classification-based approaches.* GRSG, December 2021 (online).
- *Google Earth Engine workshop.* Institute of Geosciences and Geography, Martin Luther University Halle-Wittenberg, Germany, September 2021.
- *Nitrogen dioxide (NO2) and COVID-19.* Online policy dialogue on "Linkages between air pollution and COVID-19", European Policy Centre (EPC) | Sustainable Prosperity for Europe, June 2020.
- *Development of harmonized soil spectral libraries: assessment of the influence of different laboratory set-ups and humidity conditions.* WHISPERS Conference, 2018.
- *Generating DEM from HRS data to improve classical spectral mapping.* GRSG, December 2017, Lisbon, Portugal.
- *Spatial hyper-depth spectral analysis using the VNIR–SWIR region for innovative soil classification.* GRSG, December 2016, London, United Kingdom.
- *Soil spectral library.* GEO-CRADLE Conference (EU Horizon 2020), November 2016, Limassol, Cyprus.
- *Soil spectroscopy workshop (EUFAR-2).* September 2016, Potsdam, Germany.
- *Mineral mapping of Makhtesh Ramon in Israel using hyperspectral remote sensing imagery in the VNIR–SWIR and LWIR regions.* GRSG, December 2014, London, United Kingdom.
- *Soil spectroscopy as an alternative tool for modern soil mapping.* Israeli Geographical Association (IGA), December 2014, Haifa, Israel.
- *Soil spectroscopy as an alternative tool for modern soil mapping.* Deutsches Zentrum für Luft- und Raumfahrt (DLR), March 2014, Munich, Germany.
- *Generating DEM from HRS data to improve classical spectral mapping.* Students' Conference, May 2013, Jacob Blaustein Institutes for Desert Research, Ben-Gurion University of the Negev, Israel.

ACADEMIC & PROFESSIONAL AWARDS

- Scholarship for Excellence in Studies — Department of Geography and the Human Environment, Tel Aviv University.
- Scholarship for Student Apprenticeship — Tel Aviv University.
- Smaller-Winnikow Scholarship — Porter School of Environmental Studies, Tel Aviv University.

TECHNICAL SKILLS & INSTRUMENTATION

Software & programming: Claude code, Agentic AI, ML, Python, Google Earth Engine, ENVI, ArcMap, QGIS, SNAP (Sentinel Toolboxes), SPSS, Unscrambler, Ocean Data View (ODV), Tableau.

Field & laboratory instrumentation: ASD and Spectral Evolution spectrometers; Agilent 4300 Handheld FTIR; laser diffraction particle size analyser.

Airborne & spaceborne sensors: AisaES, AisaFENIX, AisaOWL, TELOPS, HySpex, Hyperion, Landsat, Sentinel, WorldView-3, and LiDAR datasets.

LANGUAGES

Hebrew (native/fluent) · English (fluent) · German (B2.1) · Spanish (advanced course, Tel Aviv University)

ADDITIONAL INFORMATION

- Additional coursework: "Marine Sedimentology", Interuniversity Institute for Marine Sciences, Eilat, Israel.
- Advanced Open Water Diver (Nitrox and Rescue Diver certified).
- Project Management Fundamentals course.
- Agentic AI course.